Welcome to Beijing!

Open Compute for Inclusive AI:
Research, Education and Affordable Deployment

Yong Hua Lin

VP of BAAI, Founder of FlagOS

About me



Yonghua Lin is the Vice President and Chief Engineer of Beijing Academy of Artificial Intelligence, responsible for AI System, and Large Model Foundation Technology research, industry and open-source ecosystem cooperation.

She is formerly the Director of IBM China Research Lab and Distinguished Engineer at IBM, she led global AI system innovation within IBM. She has been engaged in research on system architecture, cloud computing, AI systems, computer vision, and other fields for more than 20 years.

She holds over 50 global patents and has won the ACM/IEEE Best Paper awards. She was named one of the "50 Leading Female Tech Leaders in China" by Forbes in 2019.

She is member of IEEE Women in Engineering Asia Pacific Leadership Team, Chair of IEEE Global Standard for Large Model Evaluation.

Beijing Academy of Artificial Intelligence

World-Leading AI Research Institute : Non-profit research institute pioneering innovation in AI

Driving next generation AI application through frontier research

Physical World Models

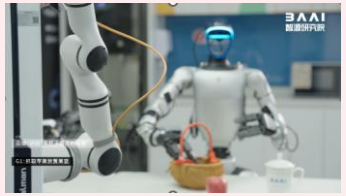


EMU3/3.5

Physis 0.1

- For Embodied AI, media industry, science discovery etc.

Embodied AI for robots



RoboBrain

ORCA

- For intelligent manufacturing, elder care, logistics and warehousing, etc.

AI for Science



BrianU

OpenComplex

- For healthcare, drug discovery, mental healthcare, etc.

Driving global innovation and collaboration through open source

Open data

116 datasets

9 millions downloads

- Open-sourced datasets of languages, vision, audio, & robotics
- Built the largest open-soured Chinese dataset CCI4.0

Open model

156 models

700 millions downloads

- Open-sourced models of multiple modalities
- BGE embedding model: the 2nd most downloaded model in 2024

Open compute

32 AI chips

FlagOS

- FlagOS: A unified open-sourced AI system software, supporting the widest range of AI chip architectures.

International collaboration

THE LINUX FOUNDATION

IEEE STANDARDS ASSOCIATION

ECLIPSE FOUNDATION

Hugging Face

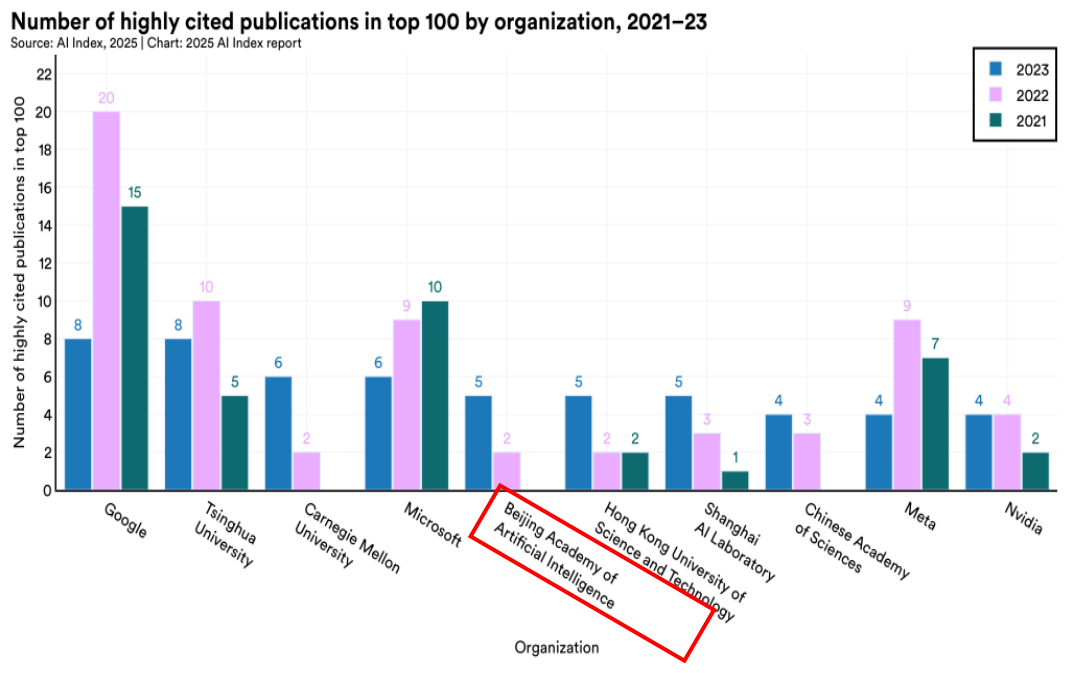
PyTorch

Beijing Academy of Artificial Intelligence

"We see three at the absolute forefront," Smith said in an interview in Tokyo with Nikkei Asia. One is **OpenAI** with Microsoft, the second is **Google**, and "the third is the **Beijing Academy of Artificial Intelligence**. "

From Brad Smith,
Vice Chairman and President, Microsoft

BAAI ranks among the world's top institutions in the number of highly cited papers.



Accelerating the development of an AI industry ecosystem

Incubated more than **10** AI startups, including one valued at over USD1.3 billion and **5** valued at more than USD 150 millions

GLM 智谱·AI

Minicpm 面壁智能

GALBOT 银河通用

RealAI 瑞莱智慧

Kimi Moonshot AI

Vidu 生数 ShengShu

星源智 XYZ Embodied AI

深言科技 DeepLang AI

智子引擎

领视智远

XIAOYU

The Reality: AI Innovation is Concentrated

70 % of global AI-grade compute are concentrated in just very few regions, locking most of the world out of frontier innovation [World Bank, 2023].

AI bridges talent gaps, but unequal compute access is building new barriers — blocking developing regions from progress in education, application, and innovation.

Three AI deployment frontiers: One shared need for choice, control and portability

As AI moves from data centers to devices and machines, freedom of deployment becomes part of practical AI sovereignty.

TOKEN FACTORY

CHOICE

Scale without lock-in

Challenge: Infrastructure lasts for years. Model architectures change in months.

Q: How can infrastructure stay useful as AI evolves?

EDGE INTELLIGENCE

CONTROL

Keep data and intelligence controllable

Challenge: AI must move closer to users and data — under tighter privacy, latency and power constraints.

Q: Can intelligence stay where the data lives?

PHYSICAL AI

PORTABILITY

Run across diverse hardware



Challenge: Real-world AI spans heterogeneous chips, devices and long product lifecycles

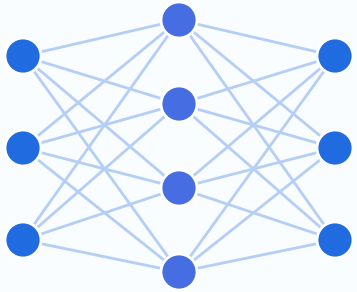
Q: Can AI stay real-time and reliable across heterogeneous hardware?

AI compute must enable choice, control, and portability.

What Is Open Compute for AI?

An open-source software stack that lets AI models run efficiently across hardware vendors, architectures and generations. —— **Develop once, run everywhere**

Open Weight Models



AI algorithms & models

Open-compute software stack

- **Model execution**
- **Operators & compilers**
- **Hardware backends**

Diverse hardware choices

CPU GPU NPU

Different vendors
Architectures and generations

More choices

Lower costs

Lower barriers

Co-create an open and shared AI ecosystem

URL: https://flagos.io Github: https://github.com/flagos-ai

FlagOS

FlagOS: Open-sourced Stack for Open Computing — Run AI Everywhere

The AI System Software Stack Supporting the Largest Number of Chip Types Worldwide

FlagOS is a unified, open-source AI system software stack that enables AI models and applications to run seamlessly across diverse hardware platforms.

20+ Chip Vendors

30+ AI Chips

There are >100 organizations contributing to this open-sourced project, including BAAI, Peking Univ., Tsinghua Univ., CSA, Alibaba, Huawei, AMD, Nvidia, etc. And FlagOS also created global collaboration with

Open-weight AI Models

Language models | Multimodal models | World models | Embodied AI

FlagOS

Unified Device Plugin for Training/Inference framework

Megatron-FL | TransformerEngine-FL | vLLM-plugin-FL | VeRL-plugin-FL | SGLang-plugin-FL | Pytorch-plugin-FL

Domain solution

FlagOS-Robo | FlagQuantums

FlagOS Skills

Deploy, code-gen, migration, benchmark

FlagOS core libraries

FlagGEMs operator library

FlagGEMs - vLLM fused-kernels

FlagGEMs, FlagBLAS, FlagSparse, FlagFFT, FlagTensor, FlagAudio

FlagScale parallel framework

FlagTree : Unified AI compiler

FlagTree: AI chips | FlagTree-CPU：ARM

FlagCX Unified comm. lib

FlagOS tools

KernelGen: operator auto-gen tool

FlagRelease: Model auto-migration tool

FlagPerf: AI chip evaluation

Diverse Compute

AI Compute Hardware (Nvidia, AMD, Huawei Ascend, ARM, etc.)

Servers of different AI chipsets



Edge devices



AI book



ARM devices



RISC-V AI chips



Quantum Compute

Quantum Cloud



Collaborate with leading AI teams in the world: Enabled Day-0 multi-chip adaption

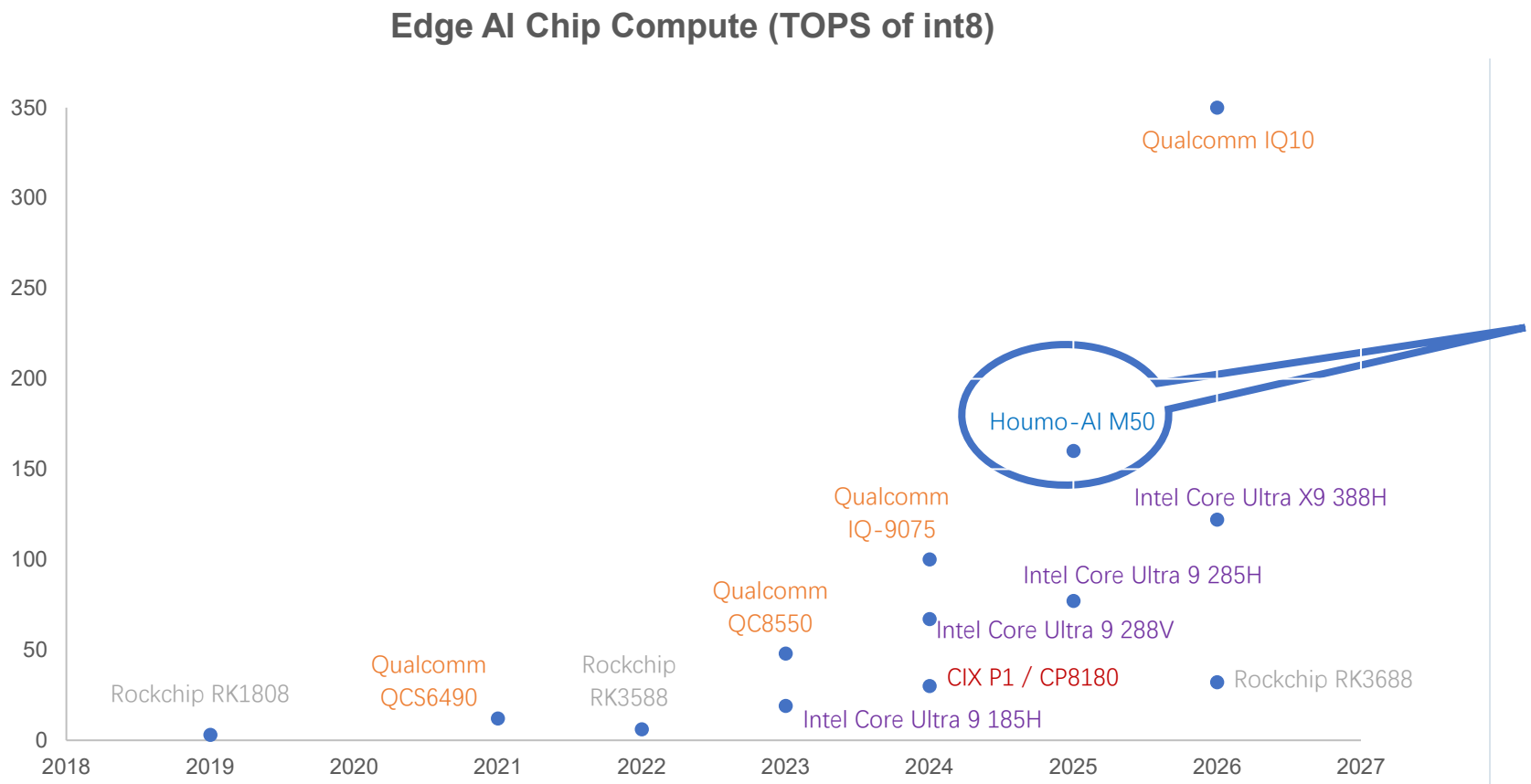
Based on FlagOS open-sourced stack, **Day-0 multi-Chip adaptation** is becoming a repeatable capability



URL for multi-chip open-model version: https://huggingface.co/FlagRelease

Value: shorten new model onboarding time and keep more infrastructure useful.

Edge AI hardware is getting stronger and more diverse



A SMALL EMBEDDED MODULE

Houmo LQ50 M.2

160

INT8 TOPS / one M50 chip

≤13W

typical module power

Up to 48GB LPDDR5

153.6GB/s memory bandwidth



Open Compute turns hardware diversity into deployment choice

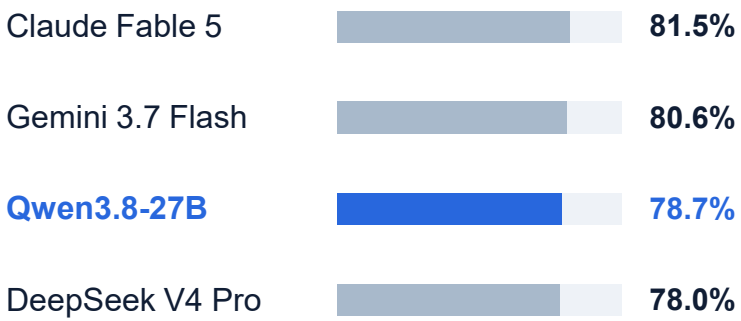
Models for Edge are getting smaller and more capable

Stronger task execution, scientific reasoning and Arabic capability — with more deployable model choices

01 ENTERPRISE TASKS

Multi-step customer service

τ²-Bench Airline • task accuracy[1]

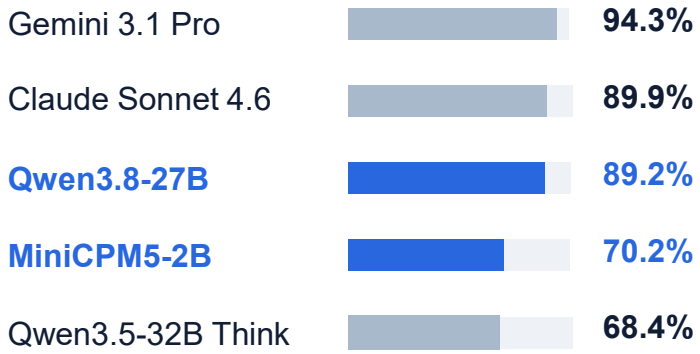


Qwen3.8-27B trails the top-ranked Claude Fable 5 by only about **2.8%**, while costing only **1/12**.

02 SCIENTIFIC REASONING

Capability at 2B–27B scale

GPQA Diamond • reported accuracy[2]



With just **2.52B** parameters, **MiniCPM5-2B** scores 70.2% on GPQA Diamond, in the range of several much larger reasoning models

03 REGIONAL LANGUAGE MODELS

ARABIC / UAE (Jan. 2026)

TII Falcon-H1 Arabic: 3B / 7B / 34B

Modern Standard Arabic + dialects
Up to 256K context

EUROPE / EU-FUNDED (Dec. 2025)

EuroLLM family: EuroMoE-2.6B-A0.6B

All 24 EU languages + 11 others[4]

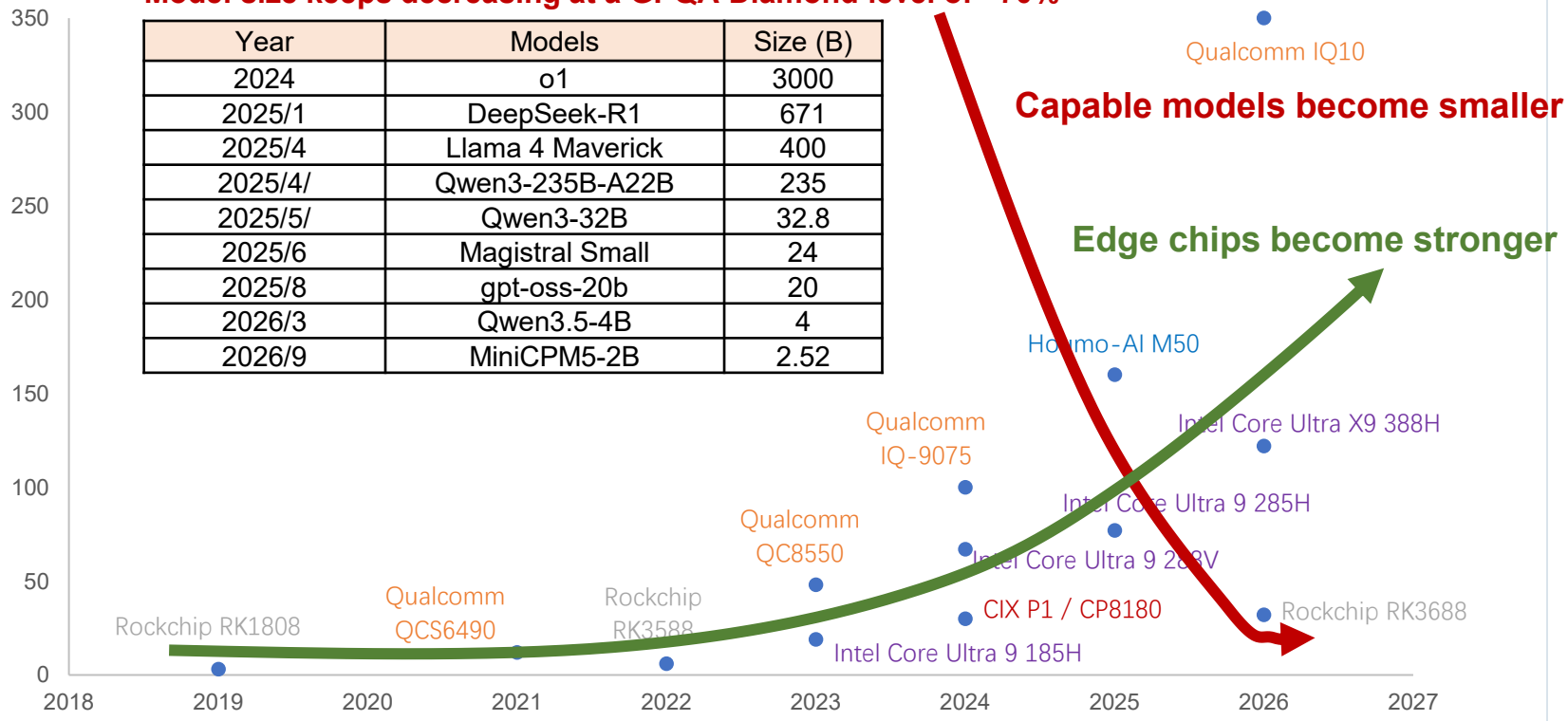
Local AI has crossed the practical threshold

Smaller capable models + stronger local compute

Edge AI Chip Compute (TOPS of int8)

Model size keeps decreasing at a GPQA-Diamond level of ~70%

Year	Models	Size (B)
2024	o1	3000
2025/1	DeepSeek-R1	671
2025/4	Llama 4 Maverick	400
2025/4/	Qwen3-235B-A22B	235
2025/5/	Qwen3-32B	32.8
2025/6	Magistral Small	24
2025/8	gpt-oss-20b	20
2026/3	Qwen3.5-4B	4
2026/9	MiniCPM5-2B	2.52



CPU-ONLY EXECUTION

FlagOS for Arm CPU

MiniCPM5-2B

Apple M5 Pro / CPU only

1,135

tokens/s prefill

90

tokens/s generation

FlagOS Supports Leading Edge AI Chips

RISC-V AI Chips

Compiler path for emerging RISC-V accelerators

- XiangShan + AME: FPGA prototype; model run completed.
- EVAS: RVV+VISA path; FlagTree compiler integration already connected.
- SpaceMiT K1/K3: edge AI CPU class; IME path and FlagGems operator-table support.
- MobileCore / ESWIN: RVV+NPU paths under technical discussion.

FlagOS value: make RISC-V ISA innovation reusable through compiler and operator infrastructure.

ARM CPU

Endpoint LLM inference without GPU/NPU dependency

- SME2 + I8MM split-GEMM path accelerates prefill by **+21–25%**.
- Mac M5 Pro CPU-only: MiniCPM5-2B W4A8 total 332 tok/s; **+39%** vs native vLLM total.
- Qwen3.8-27B: 52 tok/s total with DFlash2 path, the top record with pure ARM CPU.
- New models can be Day0-adapted onto ARM CPU.

FlagOS value: turn CPU ISA features into practical model speedups at the endpoint.

Qualcomm / Hexagon

Device AI path toward Snapdragon-class platforms

- Qwen3-0.6B W8A8 full-model forward pass runs on QCS9075 DSP.
- Pipeline: PyTorch → torch-mlir → Hexagon-MLIR → LLVM/.so → DSP runtime.
- Variable-length generate() verified; full numerical validation and FlagOS backend integration still pending.
- Target: reusable adapters, runtime lifecycle management, and optional FlagGems/Triton kernels.

FlagOS value: bridge Snapdragon / Hexagon execution into open-model frameworks.

Open Compute for Global AI Innovation

Via “Open Compute”, different types of AI compute resources from different areas can be easily used by development countries easily.

Open-source Strategy Forum in May. 2025@Paris



Area 0
AI compute resource

Area 1
AI compute resource

Area 2
AI compute resource



GDC2025，Jul. 2025@Geneva

Key technology: FlagOS
unified software ecosystem and interface that can support "open computing."

Developing countries
Users and developers

Open SDG Conference, co-organized with UN in Sep.2025@Hangzhou



Global Open Compute Initiative for AI Talent Education

In last one year, we bring it together, and make it happening

- UN Office for Digital and Emerging Technologies
- FlagOS community
- Beijing Academy of Artificial Intelligence
- Peking University, Renmin University
- Africa Union – Africa Science and Research Innovation Council
- UNDP

Global Open Compute Initiative for AI: Open Compute turns underutilized AI compute resources worldwide into accessible infrastructure for developing regions, enabling more inclusive global AI innovation.

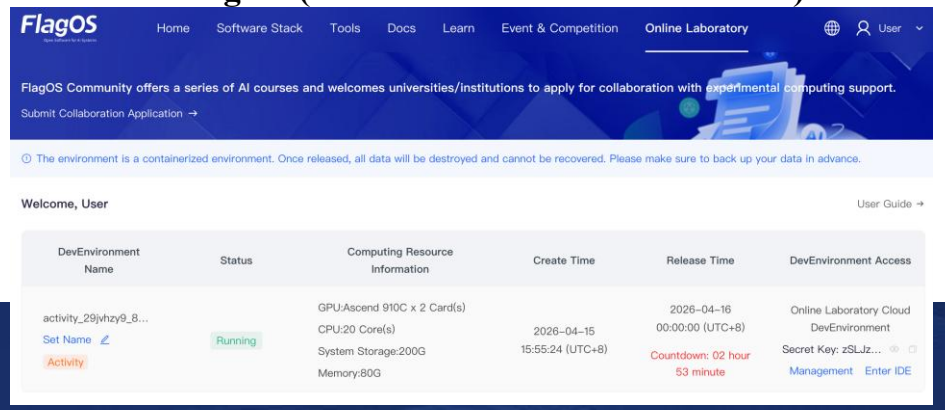


The first program: China-Africa AI Talent Development Program

BAAI, AU-ASRIC and Peking Univ. together launched the program and invited 11 teachers from different universities from 10 Africa countries. (Apr.15~Apr.30, 2026)



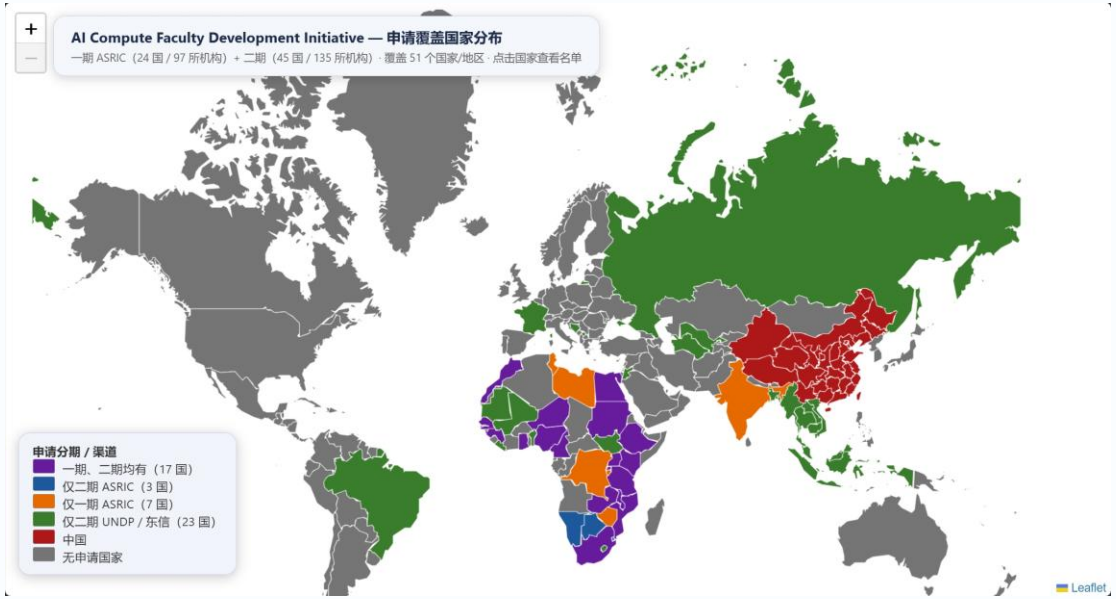
"Open Compute – online lab" powered by FlagOS (>100 AI devices from 3 vendors)



AI Compute Faculty Development Initiative (2nd Cohort)

Expanding from Africa to Asia, South America, Europe, and Beyond (October 2026)

In collaboration with the African Union–African Scientific, Research and Innovation Council, Peking University, Renmin University, the United Nations Development Programme, and other partners, BAAI will launch the second cohort of Open Compute–based AI faculty training in October 2026. Faculty members from more than 50 countries and over 100 universities have already applied to participate.



Seed Teacher Model: Plant the knowledge here. Grow the future back home.

- **Access** all teaching materials and online compute environment
- **Launch** your AI course in Africa
- **Co-create** AI innovation with teams in China

A Three-Stage Journey for Global AI Education

LEARN: Education

Open courses and hands-on labs can turn curiosity into capability and grow the next generation of AI talent.



AI Edge Technology training

DEPLOY: AI Edge Solution

Bring open models onto supported hardware so local ideas can become practical, affordable edge AI solutions.



AI4Good Research Workshop (Oct.15)

IMPACT: AI for Good Research Enablement

Accessible compute and open collaboration can help researchers turn local challenges into globally meaningful breakthroughs.



Designed as a scalable pathway from learning to deployment to global-impact research.

Three Collaboration Pillars

1 University Student Education

FlagOS-OpenCompute AI Education adapted for University student education

2 Edge AI on Supported Hardware

Prototype edge AI solutions on hardware already supported by the FlagOS-OpenCompute ecosystem.

3 Open Compute for AI-for-Good

Selected low-cost compute support for post-training, adaptation, and demonstrations with clear local impact.

Principles for this program

Respect Our Differences and Cultural Diversity

We come from diverse backgrounds. Let's stay open and respectful of each other's cultures and customs.

Adherence to Guidance

Respect local regulations and training guidelines. Use the internet appropriately to support a positive learning environment for everyone. Feel free to share your happiness on internet!

Learn with Commitment

We encourage you to dive into the course, share your voice in every session, and never hesitate to ask for help — your success is our priority.

Thanks to Our Team Members



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Start Here, Together, Let's Build an Inclusive AI Future

“When we share knowledge openly, when we break down technology barriers, when we invest in people everywhere—we unlock the full potential of AI for humanity.”